



Optimized for Routine: Highly sensitive fluorescent Telomeric Repeat Amplification Protocol (f-TRAP)

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Abstract

The strict suppression of telomerase activity (TA) in terminally differentiated human cells causes a shortening of the chromosome ends after each cell division. This tumor suppression surveillance mechanism is associated with a limited number of cell divisions known as Hayflick limit. Here we present an optimized protocol for measuring TA that combines a fluorescently labeled bait primer and polymerase chain reaction (PCR) amplification with analytical capillary electrophoresis (CE) to achieve a detection limit of one telomerase-positive cell per ten thousand negative cells. In research laboratories today, a broad panel of TRAP assay protocols enables the assessment of the immortality of newly generated cell lines or the unambiguous evaluation of the reprogramming of induced pluripotent stem cells (iPSCs). The present f-TRAP protocol, optimized for routine use, enables fast ad hoc application for single measurements up to a high throughput of mass samples using a triplicate approach of different lysate concentrations. Final CE analysis facilitates standardized data processing and storage for documentation of results and could make f-TRAP a useful assay in research and clinical oncology.

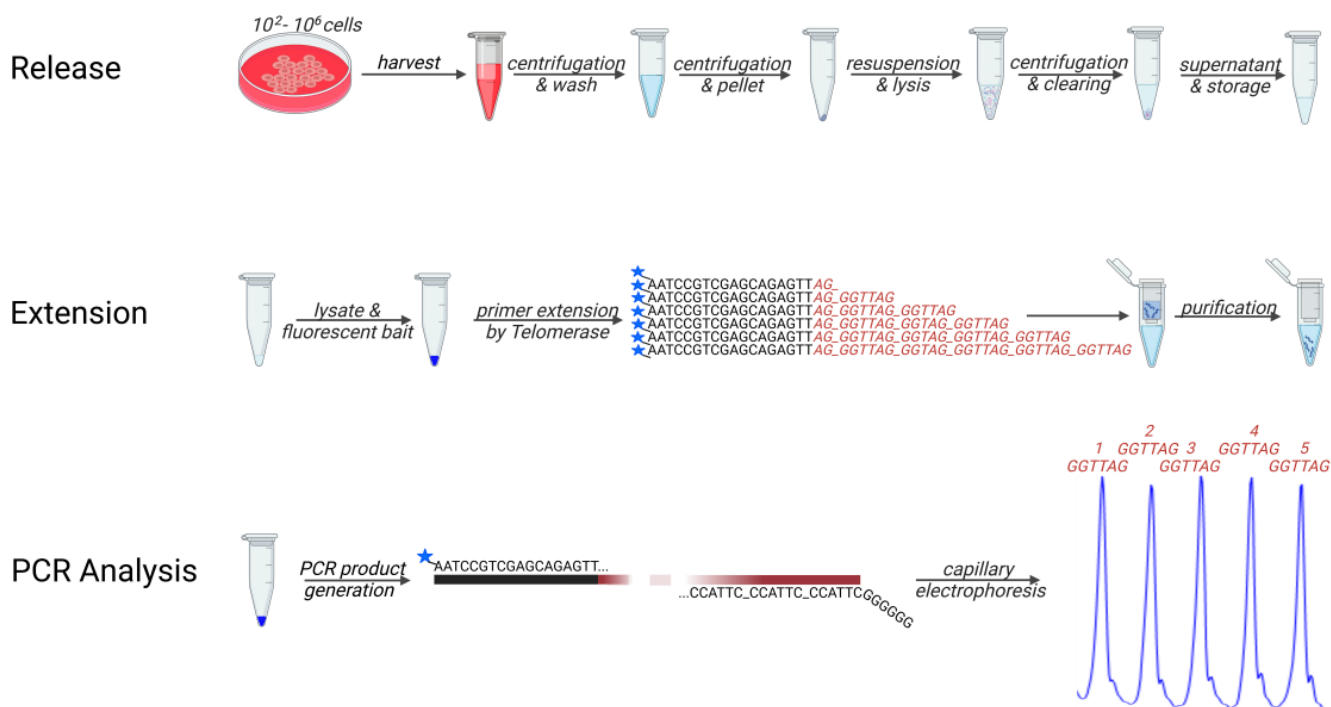


Fig. 1) Graphical display and schematic workflow overview of f-TRAP assay.

The f-TRAP assay comprises separate steps that can be performed at different times and independently of each other. Release: Generation of a protein lysate, purification of the supernatant by centrifugation and storage of the ready-to-use supernatant at -80°C . Extension: Elongation of the telomere bait primer and subsequent solid-phase silica purification of the telomerase products using spin columns to remove lysate proteins. PCR and analysis: Amplification of the extended primer products and subsequent analysis by CE.



Guidelines

Background: hTERT maintains telomere length of chromosomes in germ line, stem cells and in immortal cell lines. The telomerase ribonucleoprotein is very sensitive to RNases and temperatures! Precautions are therefore required to prevent PCR and RNase contamination. Telomerase activity of cells can be measured by the presented f-TRAP assay. Cell lines lacking telomerase activity such as U2OS and cell lines with active telomerase such as HELA should be used as negative and positive control, respectively.

Materials & Devices

- Trapeze 1x CHAPS Lysis Buffer, Merck Cat. No. S7705 (Merck KGaA, Darmstadt, Germany).
- Pierce Coomassie Plus (Thermo Scientific, Cat. No. 23238) for Bradford assay to determine the protein concentration of lysates.
- Oligo Clean & Concentrator, Zymo Research Cat. No. D4061 (Irvine, USA).
- TaKaRa Taq Hot Start Version, Takara Bio Inc. Cat. No. R007A (Shiga, Japan).
- Oligonucleotides by Eurofins (Munich, Germany).
TS (bait): 5'-(D4)-AATCCGTCGAGCAGAGTT (D4 is a WellRed fluorescent label)
R-oriACX : 5'-CTTACCCTTACCCTTACCCTAACC
- Capillary Electrophoresis: All reagents, chemicals, hardware and software from AB SCIEX Germany GmbH (Darmstadt, Germany).

Safety Warnings

⚠ Cellular inhibitors of the assay are bypassed by triplicate approaches at different protein concentrations, providing rapid results of telomerase activity with low failure rates. Pierce Coomassie Plus (Thermo Scientific, Cat. No. 23238) was used in a Bradford assay to determine the protein concentration of lysates. Depending on cell lines used concentrations were generally between 1.0 and 3.5 µg/µl protein. Triplicate aliquots of 10, 5, and 1 µl with equivalents for 9×10^4 , 4.5×10^4 and 9×10^3 cells, respectively, and a heat-inactivated equivalent of 4.5×10^4 cells were subjected to an extension reaction in a final volume of 50 µl.

Before Starting

The telomerase bait and forward TS primer is labeled at the 5'-end with the fluorophore D4 [WellRED-Oligos, Eurofins, Munich, Germany] in order to achieve a high extinction coefficient, which absorbs at 670 nm. Other fluorophores can also be used for labeling, provided the capillary electrophoresis is equipped with suitable filters.



1 h

Lysate Preparation

- 1 Count and pellet 1×10^6 cells of best viability and place in a 1,5 ml reaction cap.
- 2 Wash cells 1x using PBS at room temperature.
- 3 Dissolve cell pellets thoroughly in 50 μ l TRAPeze® 1X CHAPS lysis buffer and incubate for 30 min on wet ice.
- 4 Centrifuge lysates for 20 min at 12.000 g at 4 °C.
- 5 Save 45 μ l supernatant in a fresh cap and add 45 μ l sterile water, store at -80°C or use directly.
- 6 Pierce Coomassie Plus (Thermo Scientific, Cat. No. 23238) is used to determine the protein concentration of lysates (Bradford Assay). Depending on cell lines used concentrations were generally between 1.0 and 3.5 μ g/ μ l protein. Triplicate aliquots of 10, 5, and 1 μ l with equivalents for 9×10^4 , 4.5×10^4 and 9×10^3 cells, respectively, as well as a heat-inactivated equivalent of 4.5×10^4 cells were subjected to an extension reaction in a final volume of 50 μ l).

1 h

Extension Reaction

- 7 Cells from different tissues can differ in the quantity and quality of inhibitors for the TRAP assay. Reliable results are usually obtained with triplicate preparations and 1:10, 1:50 and 1:100 dilutions of the lysate.
- 8 An aliquot of 15 μ l out of 90 μ l supernatant is used as negative control by heat inactivation at 85°C for 10 min.
- 9 A triplicate reaction using 10 μ l lysate (corresponds to 90,000 cells), 5 μ l lysate (45,000 cells) and 1 μ l lysate (9,000 cells) guarantees a reliable result.
- 10 Pipette 5 μ l of 10xTRAP-buffer (200 mM Tris-HCL pH8,3; 15 mM MgCl₂, 630 mM KCl₂; 0,005% Tween20; 10 mM EGTA) into 0,2 ml PCR caps or cap stripes for each reaction.
- 11 Add 5 μ l of 10 mM dNTP (Takara) and 5 μ l of 1 μ M fluorescent TS primer (bait).
- 12 Add per triple set 10, 5 and 1 μ l lysate and 10 μ l of heat-inactivated lysate to the control. Fill up to a final 50 μ l volume with sterile water.
- 13 Mix thoroughly, centrifuge shortly and incubate at 30 °C for 60 min in an PCR device.
- 14 Inactivate reactions for 5 min at 94°C and keep at 4°C.



20 m

Purification of Extension Products

- 15 Add to each extension reaction 100 µl oligo binding buffer of Oligo Clean & Concentrator Kit.
- 16 Add 400 µl fresh ethanol (98%) to each extension reaction.
- 17 Pipet the total 550 µl to the column and centrifuge at 15 000 g for 30 sec.
- 18 Wash once using 750 µl wash buffer, centrifuge at 15 000 g for 60 sec.
- 19 Pipet 25 µl of sterile water to the bound extension products and incubate for 1 min.
- 20 Eluate extension products by centrifugation at 15 000 g for 30 seconds.
- 21 Keep the eluates on wet ice.

3 h

PCR Amplification

- 22 Place 20 µl of eluate into 0,2 PCR caps or cap stripes.
- 23 Prepare PCR reaction mix for n + 1 sample (2,5 µl 10X Takara buffer, 1 µl dNTP, 1 µl 10µM R-oriACX-Tel primer, 0,2 µl Takara Taq polymerase (1 U), 0,3 µl sterile water.
- 24 Add 5µl PCR reaction mix to each eluate, mix thoroughly and spin down for 10 sec.
- 25 f-TRAP PCR cyclers program:

step1: 60 sec 95°C (1 cycle),
step 2: 30 sec 94 °C, 45 sec 56 °C, 45 sec 70 °C (10 cycles),
step 3: 30 sec 90 °C, 45 sec 56 °C, 45 sec 70 °C (20 cycles),
step 4: 40 min 56 °C, ~ RT.

1 h

Analysis using Capillary Electrophoresis (CE)

- 26 Prepare 24 µl loading mix for n + 1 sample by placing 23,5 sample loading solution (SLS) and 0,5 µl of a 600 bp size standard (AB Sciex, Darmstadt, Germany).
- 27 Add 1 µl PCR reaction of sample triplicates to device-specific loading plates for CE.

- 28 Use device-specific running condition of CE for fragment analysis between 70 and 600 nucleotide length.
- 29 In general, capillary electrophoreses provide software that enables DNA fragment analysis in order to automatically evaluate the PCR result. Depending on the manufacturer, the analysis parameters are set via dialog boxes that can be saved so that the auto-run function can be activated with a single mouse click. The analysis settings include options for detecting telomeric fragment sizes of 6 bp lengths and for correcting common chemical artifacts (pull-up, peak saturation, spike removal).
- 30 For quick access and long-term documentation, evaluated f-TRAP electropherograms can be easily and securely stored in a database.

Protocol References

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